

The Typical Embedded System

Communication Interface

- ✓ Communication interface is essential for communicating with various subsystems of the embedded system and with the external world
- ✓ For an embedded product, the communication interface can be viewed in two different perspectives; namely; Device/board level communication interface (Onboard Communication Interface) and Product level communication interface (External Communication Interface)
- ✓ Embedded product is a combination of different types of components (chips/devices) arranged on a Printed Circuit Board (PCB). The communication channel which interconnects the various components within an embedded product is referred as Device/board level communication interface (Onboard Communication Interface)
- ✓ Serial interfaces like I2C, SPI, UART, 1-Wire etc and Parallel bus interface are examples of 'Onboard Communication Interface'
- ✓ The 'Product level communication interface' (External Communication Interface) is responsible for data transfer between the embedded system and other devices or modules
- ✓ The external communication interface can be either wired media or wireless media and it can be a serial or parallel interface. Infrared (IR), Bluetooth (BT), Wireless LAN (Wi-Fi), Radio Frequency waves (RF), GPRS etc are examples for wireless communication interface
- ✓ RS-232C/RS-422/RS 485, USB, Ethernet (TCP-IP), IEEE 1394 port, Parallel port, CF-II Slot, SDIO, PCMCIA etc are examples for wired interfaces

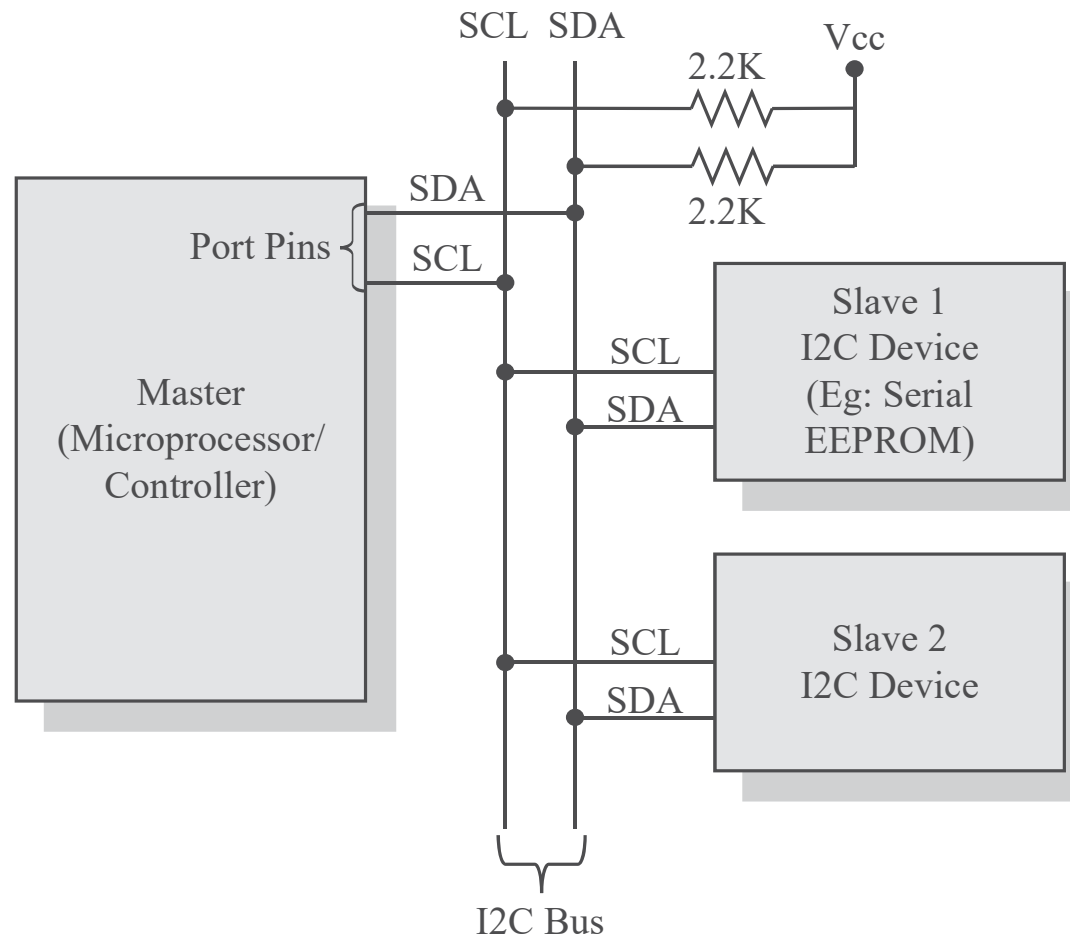
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On-board Communication Interface - I2C

- ✓ Inter Integrated Circuit Bus (I2C - Pronounced 'I square C') is a synchronous bi-directional half duplex (one-directional communication at a given point of time) two wire serial interface bus
- ✓ The concept of I2C bus was developed by 'Philips Semiconductors' in the early 1980's. The original intention of I2C was to provide an easy way of connection between a microprocessor/microcontroller system and the peripheral chips in Television sets
- ✓ The I2C bus is comprised of two bus lines, namely; Serial Clock – SCL and Serial Data – SDA. SCL line is responsible for generating synchronization clock pulses and SDA is responsible for transmitting the serial data across devices.
- ✓ I2C bus is a shared bus system to which many number of I2C devices can be connected. Devices connected to the I2C bus can act as either 'Master' device or 'Slave' device
- ✓ The 'Master' device is responsible for controlling the communication by initiating/terminating data transfer, sending data and generating necessary synchronization clock pulses
- ✓ 'Slave' devices wait for the commands from the master and respond upon receiving the commands
- ✓ 'Master' and 'Slave' devices can act as either transmitter or receiver
- ✓ Regardless whether a master is acting as transmitter or receiver, the synchronization clock signal is generated by the 'Master' device only
- ✓ I2C supports multi masters on the same bus

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On-board Communication Interface - I2C



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On-board Communication Interface - I2C

The sequence of operation for communicating with an I2C slave device is:

1. Master device pulls the clock line (SCL) of the bus to 'HIGH'
2. Master device pulls the data line (SDA) 'LOW', when the SCL line is at logic 'HIGH' (This is the 'Start' condition for data transfer)
3. Master sends the address (7 bit or 10 bit wide) of the 'Slave' device to which it wants to communicate, over the SDA line. Clock pulses are generated at the SCL line for synchronizing the bit reception by the slave device. The MSB of the data is always transmitted first. The data in the bus is valid during the 'HIGH' period of the clock signal
4. Master sends the Read or Write bit (Bit value = 1 Read Operation; Bit value = 0 Write Operation) according to the requirement
5. Master waits for the acknowledgement bit from the slave device whose address is sent on the bus along with the Read/Write operation command. Slave devices connected to the bus compares the address received with the address assigned to them
6. The Slave device with the address requested by the master device responds by sending an acknowledge bit (Bit value =1) over the SDA line
7. Upon receiving the acknowledge bit, master sends the 8bit data to the slave device over SDA line, if the requested operation is 'Write to device'. If the requested operation is 'Read from device', the slave device sends data to the master over the SDA line
8. Master waits for the acknowledgement bit from the device upon byte transfer complete for a write operation and sends an acknowledge bit to the slave device for a read operation
9. Master terminates the transfer by pulling the SDA line 'HIGH' when the clock line SCL is at logic 'HIGH' (Indicating the 'STOP' condition)

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On-board Communication Interface – Serial Peripheral Interface (SPI) Bus

The Serial Peripheral Interface Bus (SPI) is a synchronous bi-directional full duplex four wire serial interface bus. The concept of SPI is introduced by Motorola. SPI is a single master multi-slave system. It is possible to have a system where more than one SPI device can be master, provided the condition only one master device is active at any given point of time, is satisfied. SPI requires four signal lines for communication. They are:

Master Out Slave In (MOSI): Signal line carrying the data from master to slave device.
It is also known as Slave Input/Slave Data In (SI/SDI)

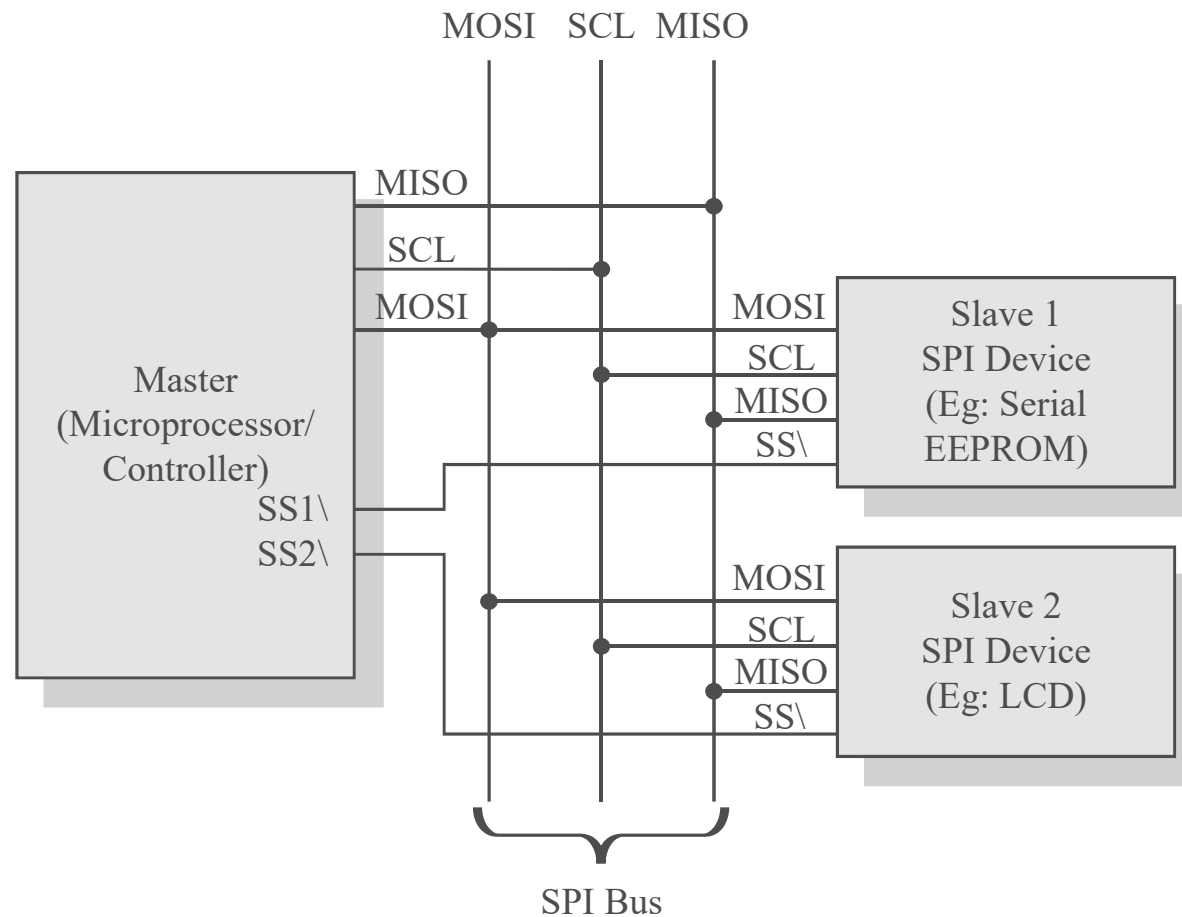
Master In Slave Out (MISO): Signal line carrying the data from slave to master device.
It is also known as Slave Output (SO/SDO)

Serial Clock (SCLK): Signal line carrying the clock signals

Slave Select (SS): Signal line for slave device select. It is an active low signal

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On-board Communication Interface – Serial Peripheral Interface (SPI) Bus



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On-board Communication Interface – Serial Peripheral Interface (SPI) Bus

- ✓ The master device is responsible for generating the clock signal. Master device selects the required slave device by asserting the corresponding slave device's slave select signal 'LOW'. The data out line (MISO) of all the slave devices when not selected floats at high impedance state
- ✓ The serial data transmission through SPI Bus is fully configurable. SPI devices contain certain set of registers for holding these configurations. The Serial Peripheral Control Register holds the various configuration parameters like master/slave selection for the device, baudrate selection for communication, clock signal control etc. The status register holds the status of various conditions for transmission and reception.
- ✓ SPI works on the principle of 'Shift Register'. The master and slave devices contain a special shift register for the data to transmit or receive. The size of the shift register is device dependent. Normally it is a multiple of 8. During transmission from the master to slave, the data in the master's shift register is shifted out to the MOSI pin and it enters the shift register of the slave device through the MOSI pin of the slave device. At the same time the shifted out data bit from the slave device's shift register enters the shift register of the master device through MISO pin

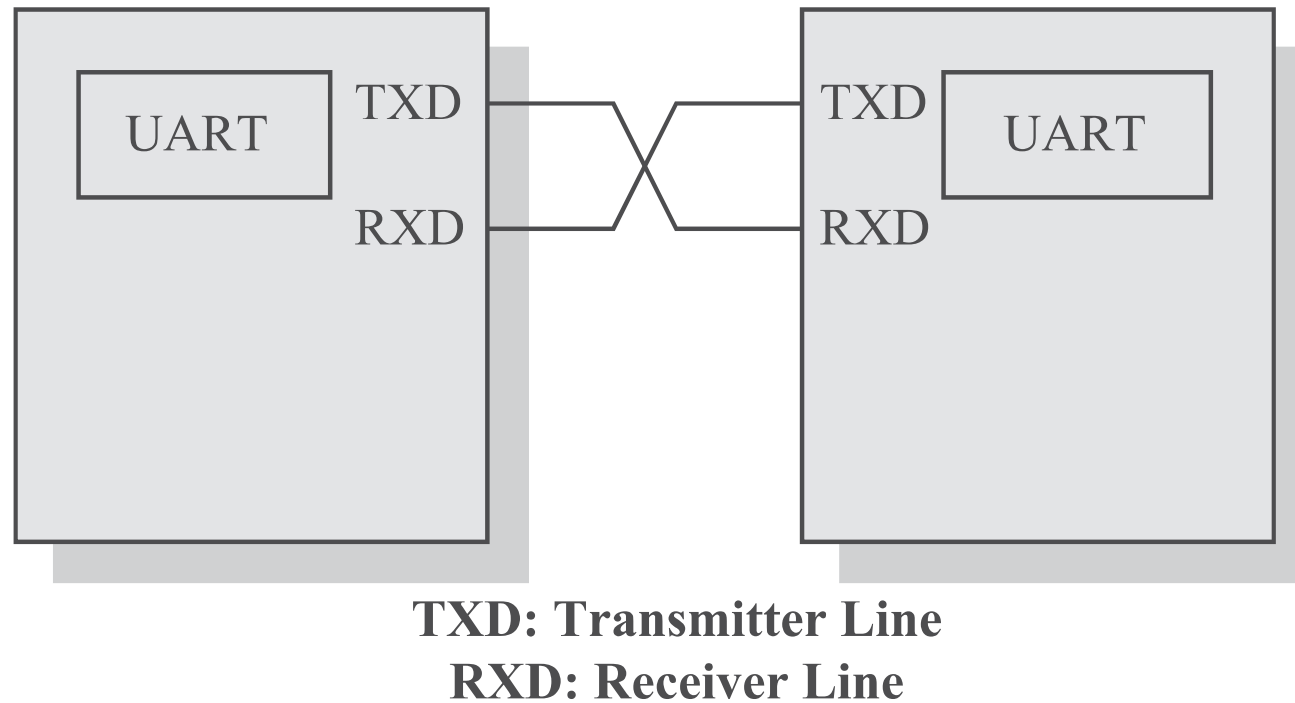
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On-board Communication Interface – Universal Asynchronous Receiver Transmitter (UART)

- ✓ Universal Asynchronous Receiver Transmitter (UART) based data transmission is an asynchronous form of serial data transmission
- ✓ The serial communication settings (Baudrate, No. of bits per byte, parity, No. of start bits and stop bit and flow control) for both transmitter and receiver should be set as identical
- ✓ The start and stop of communication is indicated through inserting special bits in the data stream
- ✓ While sending a byte of data, a start bit is added first and a stop bit is added at the end of the bit stream. The least significant bit of the data byte follows the start bit.
- ✓ The 'Start' bit informs the receiver that a data byte is about to arrive. The receiver device starts polling its 'receive line' as per the baudrate settings
- ✓ If parity is enabled for communication, the UART of the transmitting device adds a parity bit
- ✓ The UART of the receiving device calculates the parity of the bits received and compares it with the received parity bit for error checking
- ✓ The UART of the receiving device discards the 'Start', 'Stop' and 'Parity' bit from the received bit stream and converts the received serial bit data to a word

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On-board Communication Interface – Universal Asynchronous Receiver Transmitter (UART)



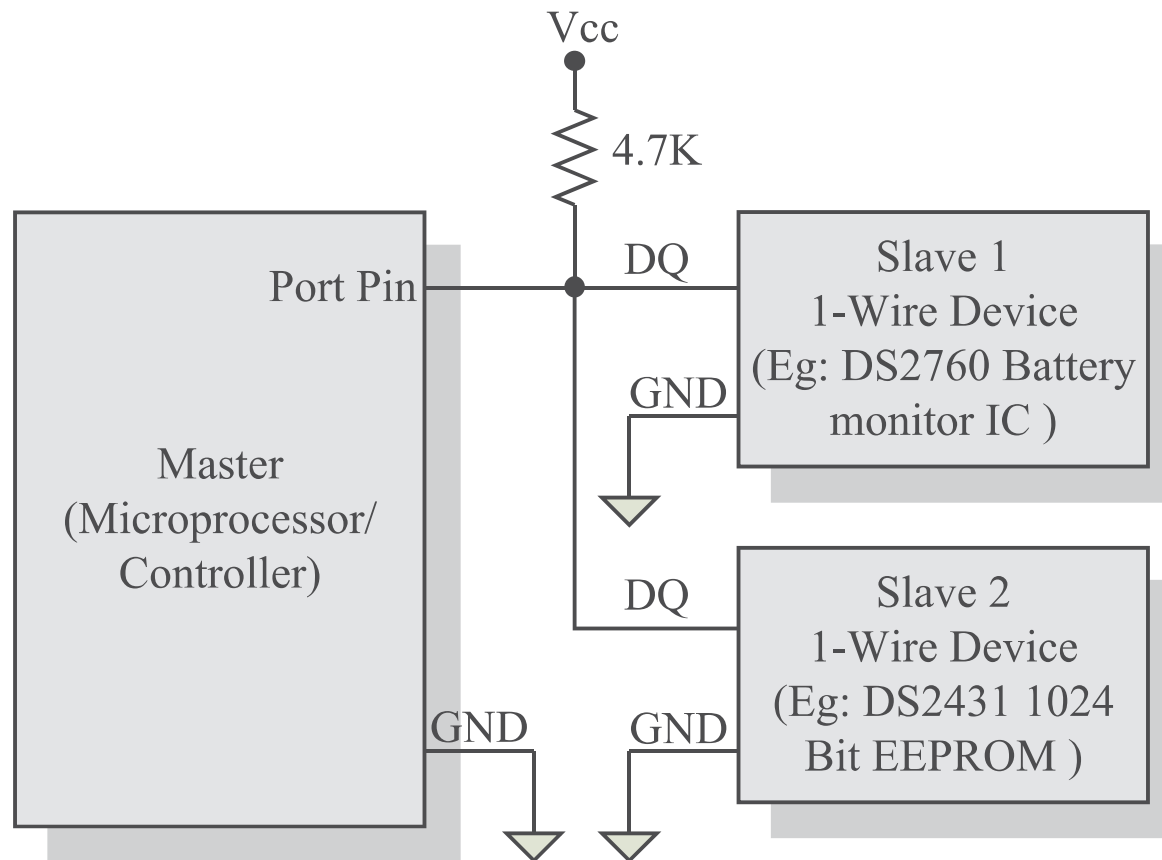
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On-board Communication Interface – 1-Wire Interface

- ✓ An asynchronous half-duplex communication protocol developed by Maxim Dallas Semiconductor (<http://www.maxim-ic.com>)
- ✓ It is also known as Dallas 1-Wire® protocol. It makes use of only a single signal line (wire) called DQ for communication and follows the master-slave communication model
- ✓ The 1-Wire interface supports a single master and one or more slave devices on the bus
- ✓ The 1-Wire is capable of carrying power to the slave device apart from carrying the signals. Slave devices incorporate internal capacitor to generate power to operate the device from the 1-Wire
- ✓ Every 1-Wire device contains a globally unique 64 bit identification number stored within it. This unique identification number can be used for addressing individual devices present in the bus in case there are multiple slave devices connected to the 1-Wire bus
- ✓ The identifier has three parts: an 8 bit family code, a 48 bit serial number and an 8 bit CRC computed from the first 56 bits

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On-board Communication Interface – 1-Wire Interface



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On-board Communication Interface – 1-Wire Interface

The sequence of operation for communicating with a 1-Wire slave device is:

1. Master device sends a 'Reset' pulse on the 1-Wire bus.
2. Slave device(s) present on the bus respond with a 'Presence' pulse.
3. Master device sends a ROM Command (Net Address Command followed by the 64 bit address of the device). This addresses the slave device(s) to which it wants to initiate a communication
4. Master device sends a read/write function command to read/write the internal memory or register of the slave device.
5. Master initiates a Read data /Write data from the device or to the device

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On-board Communication Interface – 1-Wire Interface

- ✓ All communication over the 1-Wire bus is master initiated
- ✓ The communication over the 1-Wire bus is divided into timeslots of 60 microseconds
- ✓ The 'Reset' pulse occupies 8 time slots. For starting a communication, the master asserts the reset pulse by pulling the 1-Wire bus 'LOW' for at least 8 time slots (480 μ s)
- ✓ If a 'Slave' device is present on the bus and is ready for communication it should respond to the master with a 'Presence' pulse, within 60 μ s of the release of the 'Reset' pulse by the master
- ✓ The slave device(s) responds with a 'Presence' pulse by pulling the 1-Wire bus 'LOW' for a minimum of 1 time slot (60 μ s)
- ✓ For writing a bit value of 1 on the 1-Wire bus, the bus master pulls the bus for 1 to 15 μ s and then releases the bus for the rest of the time slot
- ✓ A bit value of '0' is written on the bus by master pulling the bus for a minimum of 1 time slot (60 μ s) and a maximum of 2 time slots (120 μ s)
- ✓ To Read a bit from the slave device, the master pulls the bus 'LOW' for 1 to 15 μ s
- ✓ If the slave wants to send a bit value '1' in response to the read request from the slave, it simply releases the bus for the rest of the time slot
- ✓ If the slave wants to send a bit value '0', it pulls the bus 'LOW' for the rest of the time slot.

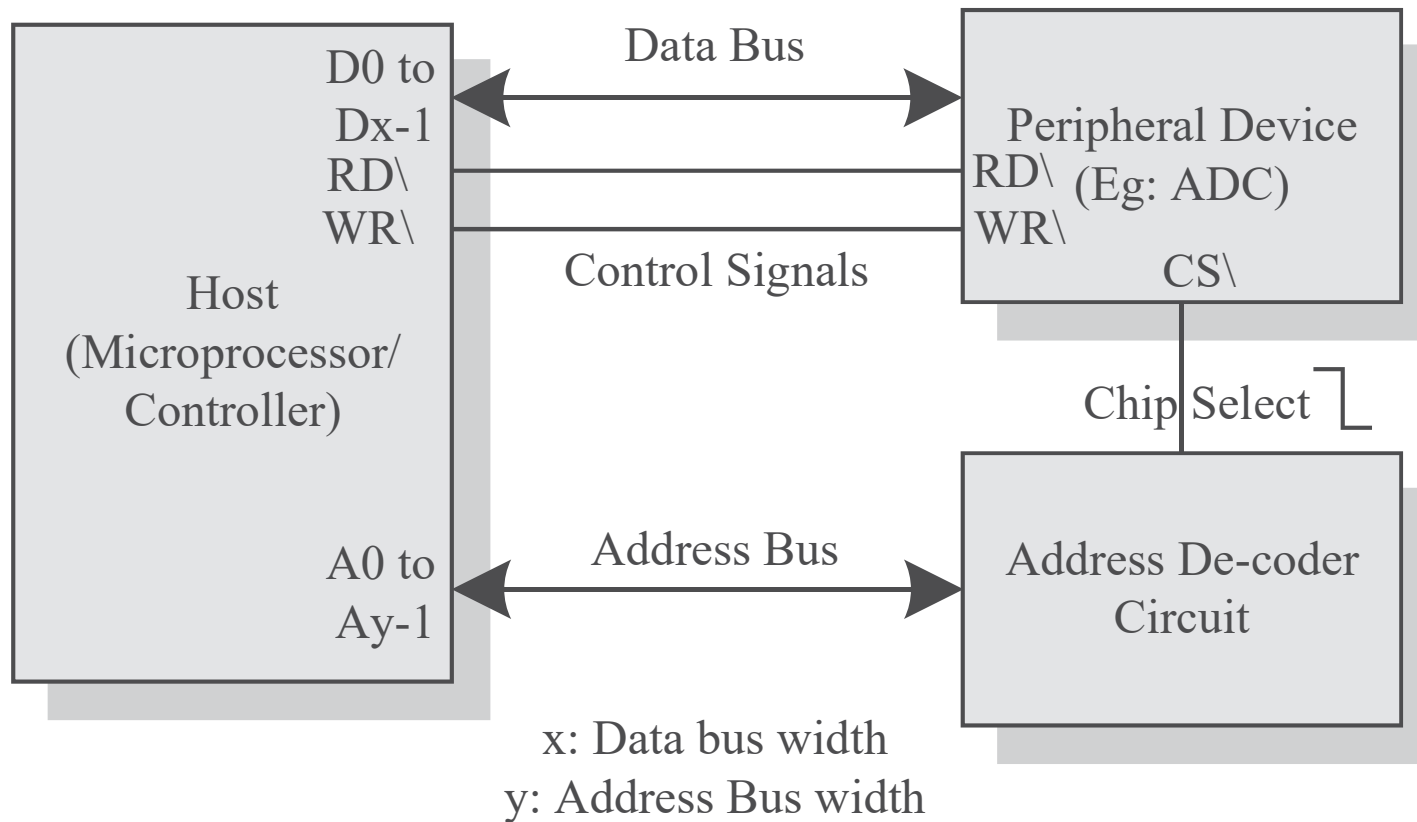
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On-board Communication Interface – Parallel Interface

- ✓ Parallel interface is normally used for communicating with peripheral devices which are memory mapped to the host of the system
- ✓ The host processor/controller of the embedded system contains a parallel bus and the device which supports parallel bus can directly connect to this bus system
- ✓ The communication through the parallel bus is controlled by the control signal interface between the device and the host
- ✓ The 'Control Signals' for communication includes 'Read/Write' signal and device select signal
- ✓ The device normally contains a device select line and the device becomes active only when this line is asserted by the host processor
- ✓ The direction of data transfer (Host to Device or Device to Host) can be controlled through the control signal lines for 'Read' and 'Write'
- ✓ Only the host processor has control over the 'Read' and 'Write' control signals

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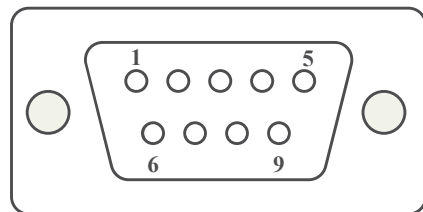
On-board Communication Interface – Parallel Interface



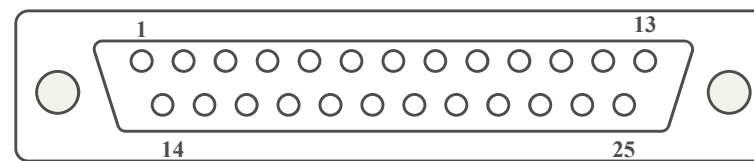
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External Communication Interface – RS-232 C & RS-485

- ✓ RS-232 C (Recommended Standard number 232, revision C from the Electronic Industry Association) is a legacy, full duplex, wired, asynchronous serial communication interface
- ✓ RS-232 extends the UART communication signals for external data communication.
- ✓ UART uses the standard TTL/CMOS logic (Logic 'High' corresponds to bit value 1 and Logic 'LOW' corresponds to bit value 0) for bit transmission whereas RS232 use the EIA standard for bit transmission. As per EIA standard, a logic '0' is represented with voltage between +3 and +25V and a logic '1' is represented with voltage between -3 and -25V. In EIA standard, logic '0' is known as 'Space' and logic '1' as 'Mark'.
- ✓ The RS232 interface define various handshaking and control signals for communication apart from the 'Transmit' and 'Receive' signal lines for data communication. RS-232 supports two different types of connectors, namely; DB-9: 9-Pin connector and DB-25: 25-Pin connector.



DB-9



DB-25

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External Communication Interface – RS-232 C & RS-485

Pin Name	Pin No: (For DB-9 Connector)	Description
TXD	3	Transmit Pin. Used for Transmitting Serial Data
RXD	2	Receive Pin. Used for Receiving serial Data
RTS	7	Request to send.
CTS	8	Clear To Send
DSR	6	Data Set ready
GND	5	Signal Ground
DCD	1	Data Carrier Detect
DTR	4	Data Terminal Ready
RI	9	Ring Indicator

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External Communication Interface – RS-232 C & RS-485

- ✓ RS-232 is a point-to-point communication interface and the devices involved in RS-232 communication are called 'Data Terminal Equipment (DTE)' and 'Data Communication Equipment (DCE)'
- ✓ If no data flow control is required, only TXD and RXD signal lines and ground line (GND) are required for data transmission and reception. The RXD pin of DCE should be connected to the TXD pin of DTE and vice versa for proper data transmission.
- ✓ If hardware data flow control is required for serial transmission, various control signal lines of the RS-232 connection are used appropriately. The control signals are implemented mainly for modem communication and some of them may be irrelevant for other type of devices
- ✓ The Request To Send (RTS) and Clear To Send (CTS) signals co-ordinate the communication between DTE and DCE. Whenever the DTE has a data to send, it activates the RTS line and if the DCE is ready to accept the data, it activates the CTS line
- ✓ The Data Terminal Ready (DTR) signal is activated by DTE when it is ready to accept data. The Data Set Ready (DSR) is activated by DCE when it is ready for establishing a communication link. DTR should be in the activated state before the activation of DSR
- ✓ The Data Carrier Detect (DCD) is used by the DCE to indicate the DTE that a good signal is being received
- ✓ Ring Indicator (RI) is a modem specific signal line for indicating an incoming call on the telephone line

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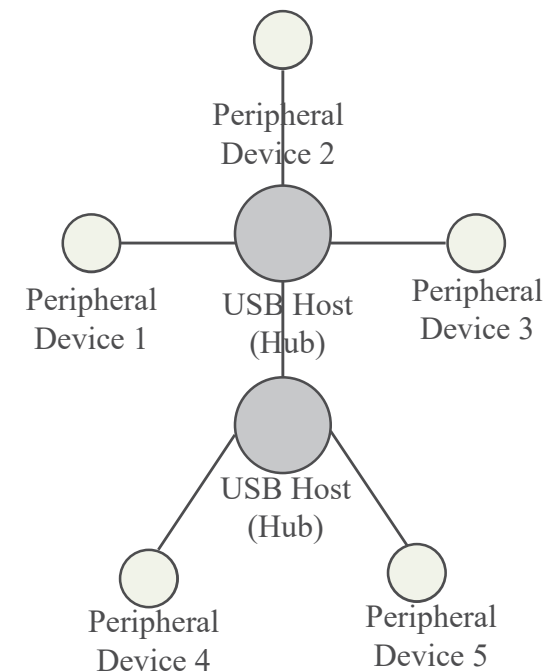
External Communication Interface – RS-232 C & RS-485

- ✓ As per the EIA standard RS-232 C supports baudrates up to 20Kbps (Upper limit 19.2Kbps) The commonly used baudrates by devices are 300bps, 1200bps, 2400bps, 9600bps, 11.52Kbps and 19.2Kbps
- ✓ The maximum operating distance supported in RS-232 communication is 50 feet at the highest supported baudrate.
- ✓ Embedded devices contain a UART for serial communication and they generate signal levels conforming to TTL/CMOS logic. A level translator IC like MAX 232 from Maxim Dallas semiconductor is used for converting the signal lines from the UART to RS-232 signal lines for communication. On the receiving side the received data is converted back to digital logic level by a converter IC. Converter chips contain converters for both transmitter and receiver
- ✓ RS-232 uses single ended data transfer and supports only point-to-point communication and not suitable for multi-drop communication
- ✓ RS-422 is another serial interface standard from EIA for differential data communication. It supports data rates up to 100Kbps and distance up to 400 ft
- ✓ RS-422 supports multi-drop communication with one transmitter device and receiver devices up to 10
- ✓ RS-485 is the enhanced version of RS-422 and it supports multi-drop communication with up to 32 transmitting devices (drivers) and 32 receiving devices on the bus. The communication between devices in the bus makes use of the 'addressing' mechanism to identify slave devices

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External Communication Interface – Universal Serial Bus (USB)

- ✓ Universal Serial Bus (USB) is a wired high speed serial bus for data communication
- ✓ The USB communication system follows a star topology with a USB host at the center and one or more USB peripheral devices/USB hosts connected to it
- ✓ A USB host can support connections up to 127, including slave peripheral devices and other USB hosts
- ✓ USB transmits data in packet format. Each data packet has a standard format. The USB communication is a host initiated one
- ✓ The USB Host contains a host controller which is responsible for controlling the data communication, including establishing connectivity with USB slave devices, packetizing and formatting the data packet. There are different standards for implementing the USB Host Control interface; namely Open Host Control Interface (OHCI) and Universal Host Control Interface (UHCI)



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External Communication Interface – Universal Serial Bus (USB)

- ✓ The Physical connection between a USB peripheral device and master device is established with a USB cable
- ✓ The USB cable supports communication distance of up to 5 meters
- ✓ The USB standard uses two different types of connectors namely ‘Type A’ and ‘Type B’ at the ends of the USB cable for connecting the USB peripheral device and host device
- ✓ ‘Type A’ connector is used for upstream connection (connection with host) and ‘Type B’ connector is used for downstream connection (connection with slave device)

Pin No:	Pin Name	Description
1	V _{BUS}	Carries power (5V)
2	D-	Differential data carrier line
3	D+	Differential data carrier line
4	GND	Ground signal line

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External Communication Interface – Universal Serial Bus (USB)

- ✓ Each USB device contains a Product ID (PID) and a Vendor ID (VID)
- ✓ The PID and VID are embedded into the USB chip by the USB device manufacturer
- ✓ The VID for a device is supplied by the USB standards forum
- ✓ PID and VID are essential for loading the drivers corresponding to a USB device for communication
- ✓ USB supports four different types of data transfers, namely; Control, Bulk, Isochronous and Interrupt
- ✓ Control transfer is used by USB system software to query, configure and issue commands to the USB device
- ✓ Bulk transfer is used for sending a block of data to a device. Bulk transfer supports error checking and correction. Transferring data to a printer is an example for bulk transfer.
- ✓ Isochronous data transfer is used for real time data communication. In Isochronous transfer, data is transmitted as streams in real time. Isochronous transfer doesn't support error checking and re-transmission of data in case of any transmission loss
- ✓ Interrupt transfer is used for transferring small amount of data. Interrupt transfer mechanism makes use of polling technique to see whether the USB device has any data to send
- ✓ The frequency of polling is determined by the USB device and it varies from 1 to 255 milliseconds. Devices like Mouse and Keyboard, which transmits fewer amounts of data, uses Interrupt transfer.

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External Communication Interface – IEEE 1394 (Firewire)

- ✓ A wired, isochronous high speed serial communication bus. It is also known as High Performance Serial Bus (HPSB)
- ✓ The research on 1394 was started by Apple Inc in 1985 and the standard for this was coined by IEEE.
- ✓ The Apple Inc's (www.apple.com) implementation of 1394 protocol is popularly known as *Firewire*.
- ✓ *i.LINK* is the 1394 implementation from Sony Corporation (www.sony.net) and *Lynx* is the implementation from Texas Instruments (www.ti.com)
- ✓ 1394 supports peer-to-peer connection and point-to-multipoint communication allowing 63 devices to be connected on the bus in a tree topology
- ✓ The 1394 standard supports a data rate of 400 to 3200Mbits/Second
- ✓ IEEE 1394 uses differential data transfer and the interface cable supports 3 types of connectors, namely; 4-pin connector, 6-pin connector (alpha connector) and 9 pin connector (beta connector)
- ✓ The 6 and 9 pin connectors carry power also to support external devices. It can supply unregulated power in the range of 24 to 30V (The Apple implementation is for battery operated devices and it can supply a voltage in the range 9 to 12V)

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External Communication Interface – IEEE 1394 (Firewire)

Pin Name	Pin No: (4 Pin Connector)	Pin No: (6 Pin Connector)	Pin No: (9 Pin Connector)	Description
Power		1	8	Unregulated DC supply. 24 to 30V
Signal Ground		2	6	Ground connection
TPB-	1	3	1	Differential Signal line for Signal Line B
TPB+	2	4	2	Differential Signal line for Signal Line B
TPA-	3	5	3	Differential Signal line for Signal Line A
TPA+	4	6	4	Differential Signal line for Signal Line A
TPA(S)			5	Shield for the differential signal line A. Normally grounded
TPB(S)			9	Shield for the differential signal line B. Normally grounded
NC			7	No connection

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External Communication Interface – IEEE 1394 (Firewire)

- ✓ The IEEE 1394 connector contains two differential data transfer lines namely A and B
- ✓ The differential lines of A are connected to B (TPA+ to TPB+ and TPA- to TPB-) and vice versa
- ✓ Unlike USB interface (Except USB OTG), IEEE 1394 doesn't require a host for communicating between devices. Example, a scanner can be directly connected to a printer for printing
- ✓ The data rate supported by 1394 is far higher than the one supported by USB2.0 interface
- ✓ 1394 is a popular communication interface for connecting embedded devices like 'Digital Camera', 'Camcorder', 'Scanners' with desktop Computers for data transfer and storage

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External Communication Interface – Infrared (IrDA)

- ✓ A serial, half duplex, line of sight based wireless technology for data communication between devices
- ✓ Infrared communication technique makes use of Infrared waves of the electromagnetic spectrum for transmitting the data
- ✓ IrDA supports point-point and point-to-multipoint communication, provided all devices involved in the communication are within the line of sight
- ✓ The typical communication range for IrDA lies in the range 10cm to 1 m
- ✓ IR supports data rates ranging from 9600bits/second to 16Mbps. Depending on the speed of data transmission IR is classified into Serial IR (SIR), Medium IR (MIR), Fast IR (FIR), Very Fast IR (VFIR) and Ultra Fast IR (UFIR)
- ✓ SIR supports transmission rates ranging from 9600bps to 115.2kbps. MIR supports data rates of 0.576Mbps and 1.152Mbps. FIR supports data rates up to 4Mbps. VFIR is designed to support high data rates up to 16Mbps. The UFIR specs are under development and it is targeting a data rate up to 100Mbps
- ✓ IrDA communication involves a transmitter unit for transmitting the data over IR and a receiver for receiving the data. Infrared Light Emitting Diode (LED) is used as the IR source for transmitter and at the receiving end a photodiode is used as the receiver

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External Communication Interface – Bluetooth

- ✓ Low cost, low power, short range wireless technology for data and voice communication
- ✓ Bluetooth operates at 2.4GHz of the Radio Frequency spectrum and uses the Frequency Hopping Spread Spectrum (FHSS) technique for communication.
- ✓ Bluetooth supports a theoretical maximum data rate of up to 1Mbps and a range of approximately 30 feet for data communication
- ✓ Bluetooth communication has two essential parts; a physical link part and a protocol part. The physical link is responsible for the physical transmission of data between devices supporting Bluetooth communication and protocol part is responsible for defining the rules of communication
- ✓ The physical link works on the Wireless principle making use of RF waves for communication
- ✓ Bluetooth enabled devices essentially contain a Bluetooth wireless radio for the transmission and reception of data
- ✓ The rules governing the Bluetooth communication is implemented in the 'Bluetooth protocol stack'. The Bluetooth communication IC holds the stack
- ✓ Each Bluetooth device will have a 48 bit unique identification number. Bluetooth communication follows packet based data transfer
- ✓ Bluetooth supports point-to-point (device to device) and point-to-multipoint (device to multiple device broadcasting) wireless communication. The point-to-point communication follows the master-slave relationship. A Bluetooth device can function as either master or slave
- ✓ A network formed with one Bluetooth device as master and more than one device as slaves is known as Piconet

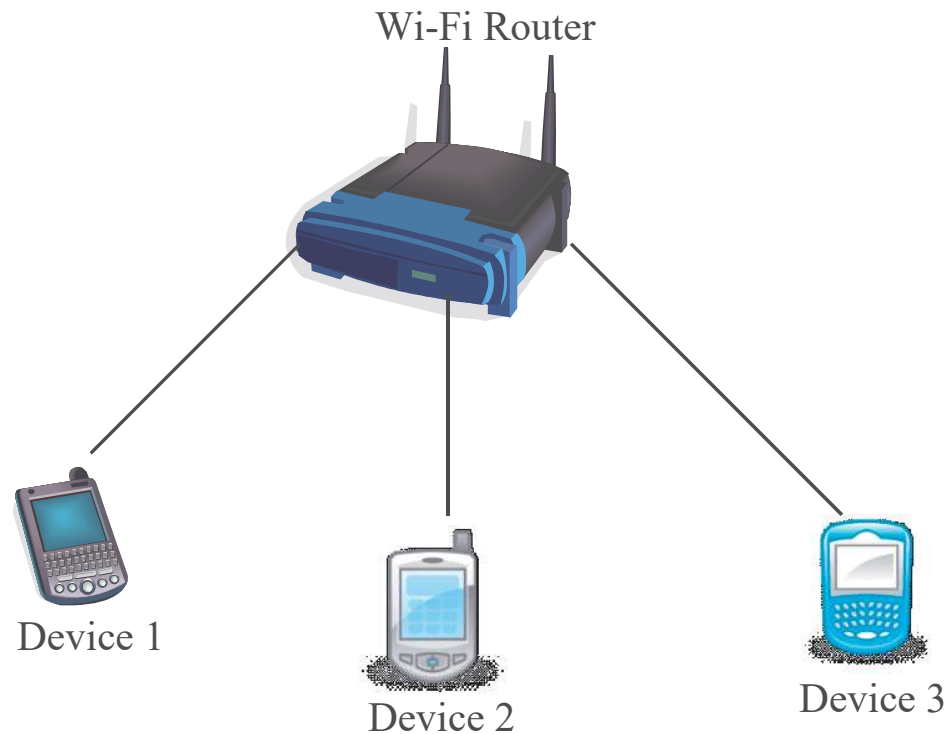
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External Communication Interface – Wi-Fi

- ✓ The popular wireless communication technique for networked communication of devices
- ✓ Wi-Fi follows the IEEE 802.11 standard
- ✓ Wi-Fi is intended for network communication and it supports Internet Protocol (IP) based communication
- ✓ Wi-Fi based communications require an intermediate agent called Wi-Fi router/Wireless Access point to manage the communications
- ✓ The Wi-Fi router is responsible for restricting the access to a network, assigning IP address to devices on the network, routing data packets to the intended devices on the network
- ✓ Wi-Fi enabled devices contain a wireless adaptor for transmitting and receiving data in the form of radio signals through an antenna
- ✓ Wi-Fi operates at 2.4GHZ or 5GHZ of radio spectrum and they co-exist with other ISM band devices like Bluetooth
- ✓ A Wi-Fi network is identified with a Service Set Identifier (SSID). A Wi-Fi device can connect to a network by selecting the SSID of the network and by providing the credentials if the network is security enabled
- ✓ Wi-Fi networks implements different security mechanisms for authentication and data transfer
- ✓ Wireless Equivalency Protocol (WEP), Wireless Protected Access (WPA) etc are some of the security mechanisms supported by Wi-Fi networks in data communication

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External Communication Interface – Wi-Fi



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External Communication Interface – ZigBee

- ✓ Low power, low cost, wireless network communication protocol based on the IEEE 802.15.4-2006 standard
- ✓ ZigBee is targeted for low power, low data rate and secure applications for Wireless Personal Area Networking (WPAN)
- ✓ The ZigBee specifications support a robust mesh network containing multiple nodes. This networking strategy makes the network reliable by permitting messages to travel through a number of different paths to get from one node to another.
- ✓ ZigBee operates worldwide at the unlicensed bands of Radio spectrum, mainly at 2.400 to 2.484 GHz, 902 to 928 MHz and 868.0 to 868.6 MHz
- ✓ ZigBee Supports an operating distance of up to 100 meters and a data rate of 20 to 250Kbps
- ✓ ZigBee is primarily targeting application areas like Home & Industrial Automation, Energy Management, Home control/security, Medical/Patient tracking, Logistics & Asset tracking and sensor networks & active RFID
- ✓ Automatic Meter Reading (AMR), smoke and detectors, wireless telemetry, HVAC control, heating control, Lighting controls, Environmental controls, etc are examples for applications which can make use of the ZigBee technology

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External Communication Interface – ZigBee

In the ZigBee terminology, each ZigBee device falls under any one of the following ZigBee device category

ZigBee Coordinator (ZC)/Network Coordinator:

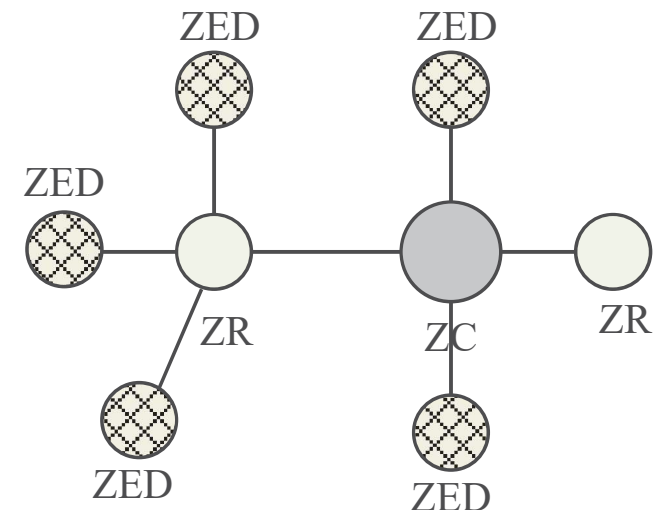
The ZigBee coordinator acts as the root of the ZigBee network. The ZC is responsible for initiating the ZigBee network and it has the capability to store information about the network

ZigBee Router (ZR)/Full function Device (FFD):

Responsible for passing information from device to another device or to another ZR

ZigBee End Device (ZED)/Reduced Function Device (RFD):

End device containing ZigBee functionality for data communication. It can talk only with a ZR or ZC and doesn't have the capability to act as a mediator for transferring data from one device to another.



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External Communication Interface – General Packet Radio Service (GPRS)

- ✓ A communication technique for transferring data over a mobile communication network like GSM
- ✓ Data is sent as packets. The transmitting device splits the data into several related packets. At the receiving end the data is re-constructed by combining the received data packets
- ✓ GPRS supports a theoretical maximum transfer rate of 171.2kbps
- ✓ In GPRS communication, the radio channel is concurrently shared between several users instead of dedicating a radio channel to a cell phone user. The GPRS communication divides the channel into 8 timeslots and transmits data over the available channel
- ✓ GPRS supports Internet Protocol (IP), Point to Point Protocol (PPP) and X.25 protocols for communication.
- ✓ GPRS is mainly used by mobile enabled embedded devices for data communication. The device should support the necessary GPRS hardware like GPRS modem and GPRS radio
- ✓ GPRS is an old technology and it is being replaced by new generation data communication techniques like EDGE, High Speed Downlink Packet Access (HSDPA) etc which offers higher bandwidths for communication